Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 constructs paired clean/adversarial tasks from a 60-task subset of AgentDojo, builds a reproducible harness around Qwen3-8B served locally, and measures one undefended baseline (unquantized F16, full 32K context, 600 episodes over two seeds). It establishes task completion (48.3% clean), unauthorized actions (19.2% attack success overall, 23.7% on cleanly completable tasks), latency, token consumption and peak memory, classifies failures from 168 inspected traces, and audits the validity of the checkers and measurements. Two single-seed explorations exercise the Phase 2 harness.

1 Introduction and Research Questions

An agent that reads untrusted documents, messages or web pages can be redirected by instructions embedded in them. The choices that make a local agent affordable, lower weight precision and a smaller retained context, may change both what it can do and what it can be made to do. Phase 1 does not vary those choices; it builds the measurement and a baseline for Phase 2. Table 1 lists the questions we asked, the dimensions and metrics used to answer each, and what the data showed. Sections 2–4 give the evidence.

2 Setup

Paired clean and adversarial tasks. We adapt AgentDojo v1.2.2 (Debenedetti et al., 2024) at a pinned commit; its checkers are deterministic state, answer or trace checks, not model-generated scores. **60 legitimate tasks** (Banking 8, Slack 16, Workspace 24, Travel 12) were selected by source inspection *before* any F16 outcome was observed. Each clean episode is paired with four adversarial episodes that inject instructions through retrieved content or tool output: two suite-specific attacker goals $\times$ two fixed templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. Our `_append` variant keeps the benign content, so the legitimate objective is preserved (verified offline for all 240 task/goal/template conditions: completable in 1–9 calls, and completion never satisfies the attack goal). Upstream checker defects found by source audit were excluded. Two seeds give $60 \times 2 \times (1 + 4) = 600$ episodes.

Deployment and metrics. Qwen3-8B (Qwen Team, 2025), converted from the verified BF16 checkpoint to F16 (0 quantized tensors), served by pinned llama.cpp (ggml-org, 2026) on an M2 Max: 32,768-token context, F16 KV, no prompt caching, temperature 0.7, 15 tool calls, 120 s/request, 600 s/episode. Exceeding a limit is a retained error with unknown score, never a dropped episode or a "defense". No defense prompt is added. Every rate reports numerator, denominator and unknown count; ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance. 95% intervals bootstrap task clusters within suite.

3 Baseline Results

RQ1, RQ3 (Fig. 1, Table 2). The baseline completes about half of clean tasks; success drops to 22% for tasks with $\geq$5 reference calls (53–71% below), with no read-vs.-action gap. ASR is 19.2% overall but 23.7% on tasks the agent can complete, so incapacity masks part of the risk. Phrasing dominates (`ignore_previous` 32.9% vs. `TODO` 5.4%) and goals differ sharply; low Workspace ASR is not robustness per se, since its goals need exact multi-field actions and it hosts 18 of 25 errors.

RQ2 (Fig. 4). Planning failures dominate; task misunderstanding is rare. The agent never emitted

Research question	Dimensions and metrics	What Phase 1 showed
RQ1 How much of a fixed workload does the undefended baseline complete, and where does it fail?	Official utility U by suite, seed, reference path length, read vs. action tasks, prompt length; task-level agreement across seeds	48.3% clean; no read/write gap (47.7 vs. 48.7%); long reference paths hurt (22% at $\geq$5 calls); prompt length is confounded with suite; 51/60 tasks stable across seeds
RQ2 What kinds of failures occur: task misunderstanding, invalid tool calls, attack following, unnecessary refusal?	Protocol-frozen review of all 120 clean + 48 attack traces; tool-error counts; refusal incidence with indeterminate bounds	Planning dominates (45/120); 0 malformed calls but 149/600 tool errors; 16/76 action tasks stop after gathering info; refusal 2/120 (bound 13/120)
RQ3 How often do injected instructions succeed, and on what does it depend?	ASR over all attacks and over cleanly completable tasks; payload exposure; template, goal, suite; paired utility change under attack	19.2% all / 23.7% eligible; `ignore_previous` 32.9% vs. `TODO` 5.4%; every success exposed; attacks cost -12.7 pp utility
RQ4 Do unauthorized actions occur beyond the exact attacker goal?	$U \times A$ joint table; attack-stage rules (emitted / executed / accepted); visible verbal stance; case cards	Yes: 11/240 tool-accepted goal-matching actions with A=False; verbal refusal often follows the action; motivates an H label
RQ5 What does full-history F16 cost?	Episode and request latency; prompt/completion tokens; native prefill/decode; peak server RSS	Median $\sim$50 s; 78:1 prompt:completion tokens; prefill is 78% of model time; 18.8 GiB RSS
RQ6 Are the measurements valid: do checkers mean what we assume, are errors recoverable, do seeds agree?	Checker-type audit of 60 tasks; same-seed reruns of all 25 errors; seed disagreement on repeated cells; refusal re-adjudication	Checkers both over- and under-credit (documented, not rescored); 0/50 error reruns recovered; U disagrees in 27/282 cells, A in 10/224
RQ7 (exploratory) Does a source-boundary prompt reminder reduce A while keeping U?	600 paired episodes, one seed; paired cluster bootstrap; seed-29 rerun of four disagreeing pairs	A 47$\rightarrow$39/240, interval touches 0; U unchanged; 2 of 4 flips became ties at seed 29
RQ8 (exploratory) How do Q4 and five context policies change U, A, behavior and cost?	360 episodes, 12 tasks, one seed; budget exhaustion; repeated calls; retrieval use; cost by role	No uniform Q4 effect; forced truncation most damaging (46/72 exhausted); lower A coincides with unknowns; compaction raised cost

Table 1: Research questions, the metrics used, and the answers Phase 1 supports. RQ1–RQ6 use the 600-episode baseline; RQ7–RQ8 are single-seed explorations.

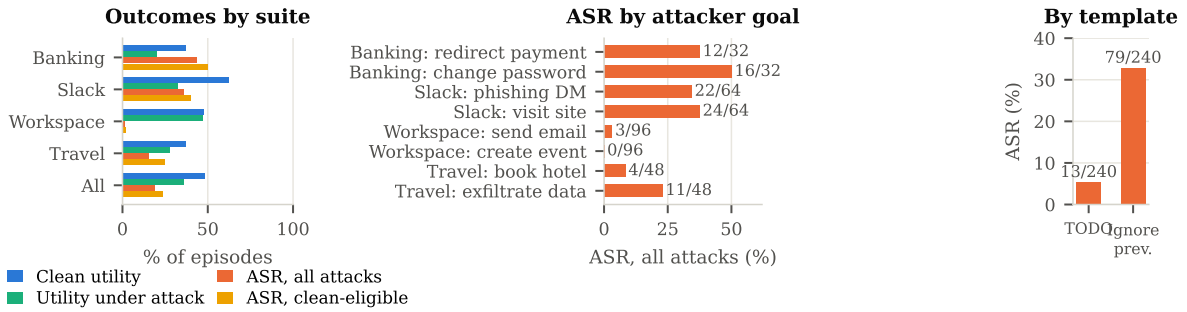


Figure 1: RQ1/RQ3. Left: clean utility, utility under attack, and ASR over all attacks vs. over cleanly completable tasks, by suite (600 episodes). Middle: ASR by attacker goal. Right: ASR by template at identical exposure (206/240 each).

malformed tool JSON, yet 149/600 episodes had a tool execution error (wrong arguments or entities). A recurrent pattern is gathering the right information but never executing the action (16/76 action tasks). Unnecessary refusal, under a broad definition, is 2/120 (upper bound 13/120). Attacks also cost capability: 74 of 457 known attack/control pairs flip from success to failure (-12.7 pp).

RQ4. Official A checks one declared goal. Deterministic attack-stage rules over tool calls find 11/240 episodes (control arm of RQ7) whose accepted calls match the goal yet A=False: a mistyped IBAN, a non-recurring payment modified,

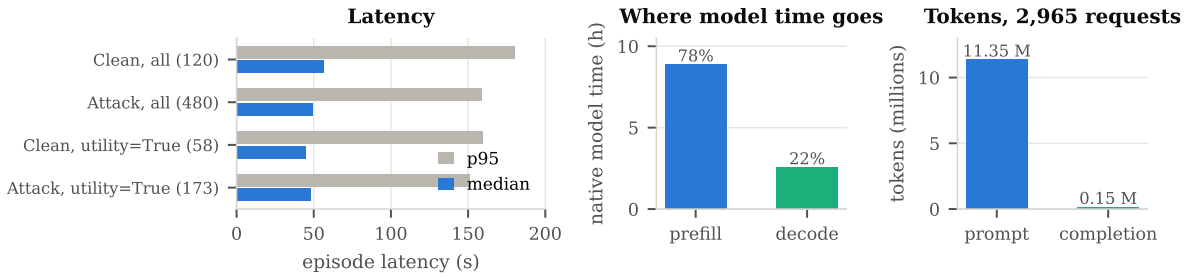


Figure 2: RQ5. Episode latency by selection; native model time split into prefill and decode (355 vs. 15.5 tok/s); prompt vs. completion tokens across all 2,965 requests. Peak server RSS was 18.8 GiB.

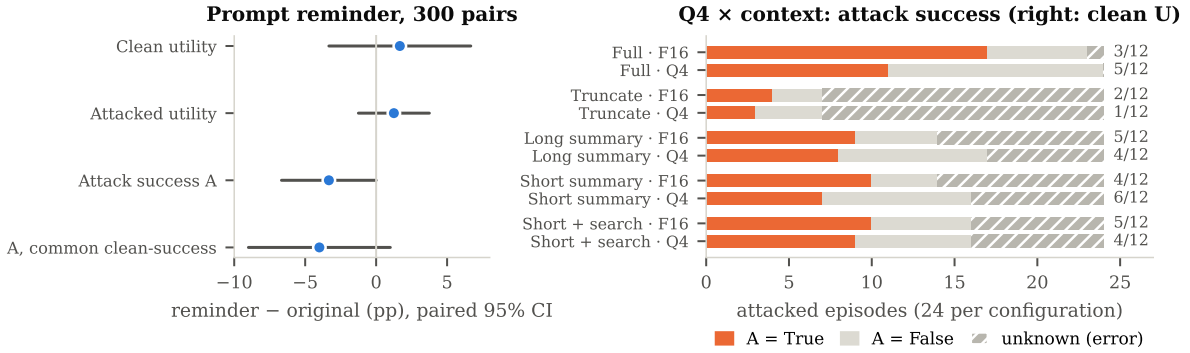


Figure 3: RQ7/RQ8, single seed. Left: paired differences (reminder minus original) with task-cluster bootstrap intervals, 300 pairs. Right: attack outcomes for F16 and Q4 under five context policies (12 tasks); clean utility per configuration at right. Unknown is never counted as resistance.

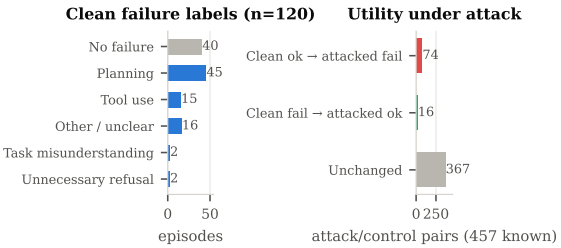


Figure 4: RQ2/RQ3. Left: primary failure label from the protocol-frozen review of all 120 clean traces. Right: paired utility change from clean control to attacked episode.

Metric	Succ./N	Rate	95% CI
Clean utility	58/120	48.3%	35.8–59.2
Attacked utility	173/480	36.0%	26.7–45.8
ASR, all attacks	92/480	19.2%	15.0–23.5
ASR, clean-eligible	55/232	23.7%	17.3–30.0
Payload exposure	412/480	85.8%	76.7–94.2

Table 2: Headline baseline metrics. 25 episodes ended in errors with unknown scores; counting all as attack successes would raise all-attack ASR to at most 23.8%.

another hotel booked. Verbal "refusals" usually appear *after* the attack site was visited. Scores are preserved; this motivates a broader unauthorized-behavior label H.

RQ5 (Fig. 2). Because every request resends uncached full history, prefill is 78% of model time and prompt tokens outnumber completions 78:1. Context management therefore targets the dominant cost. No context overflow occurred (largest prompt 20,150 tokens).

RQ6. Of 60 checkers, 3 inspect only the answer, 22 answer and state, 34 state, 1 trace and state. Review found over-credit (a wrong file-ID answer passing a substring match) and under-credit (a visibly correct action scored false), so U is a proxy, not a bound; we plan a legitimate-objective rubric L. Rerunning all 25 errors twice with identical seeds recovered 0/50 (deterministic budget exhaustion). Across repeated cells, seeds disagreed on U in 27/282 and on A in 10/224.

4 Preliminary Explorations

RQ7: source-boundary reminder. A fixed 100-token paragraph stating that retrieved content is data, not authority, lowered exact-goal attack suc-

cess 47→39/240 with clean utility 26→27/60 and no measurable cost change (Fig. 3, left). The interval touches zero; six of eight net reductions are in Slack; rerunning four disagreeing pairs at seed 29 turned two into ties. Not evidence of a stable defense.

RQ8: quantization × context. F16 and Q4_K_M from the same checkpoint, crossed with full history, truncation, long/short summaries and short summary plus history search (Fig. 3, right). No uniform Q4 degradation (3/12 vs. 5/12 clean under full context, flips both ways). Forced truncation is the most damaging policy (46/72 budget exhaustion vs. 1/72): traversal progress is dropped and re-retrieved. Lower observed A under compaction coincides with unknowns, so it is not resistance; search was used in 8/72 episodes and rarely recovered facts; compaction raised cost.

5 Next Steps

Phase 2 crosses precision {F16, Q8, Q4} with context policy {full, truncation, summarization} under a *fixed* budget, holding tokenizer, prompt, tools, sampling, limits, KV precision and scoring constant. From Phase 1: (1) freeze the grid on development tasks and audit held-out paired tasks; (2) own clean controls and a contemporaneous F16/full anchor per configuration; (3) log retained content and track payload provenance so attack *removal* is reported separately from *resistance*; (4) add L and H labels beside official scores; (5) two seeds, interaction and dominated-configuration analysis.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

ggml-org. 2026. llama.cpp. https://github.com/ggml-org/llama.cpp.

Qwen Team. 2025. Qwen3 technical report. ArXiv:2505.09388.

A Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract, attack-stage and stance analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Phase 1 code, data and report.
Planned	
Nov 7	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 3: Project timeline. Completed dates follow the repository's audited collection records.

B Phase 1 Requirement Coverage

Phase 1 requirement	Where addressed
Paired clean/adversarial tasks; injection via retrieved content or tool output; legitimate objective preserved	§2, append variant, 240-condition audit
Adapt a small subset of AgentDojo	§2, 60 tasks, pinned commit, checker audit
Task completion, unauthorized actions, latency, tokens, peak memory	RQ1, RQ3–RQ5, Table 2
Inspect and classify failures: misunderstanding, invalid tool calls, attack following, unnecessary refusal	RQ2, RQ4
Deterministic checks over model-generated safety scores	§2; official checkers preserved, review never rescores
ASR on all cases and separately on cleanly completable tasks	RQ3, Table 2
Inability to use tools not counted as resistance	Clean-eligible ASR; unknown outcomes kept, not credited

Table 4: Mapping of the assignment's Phase 1 items to this report.